

Unit:2

Operating System Structure

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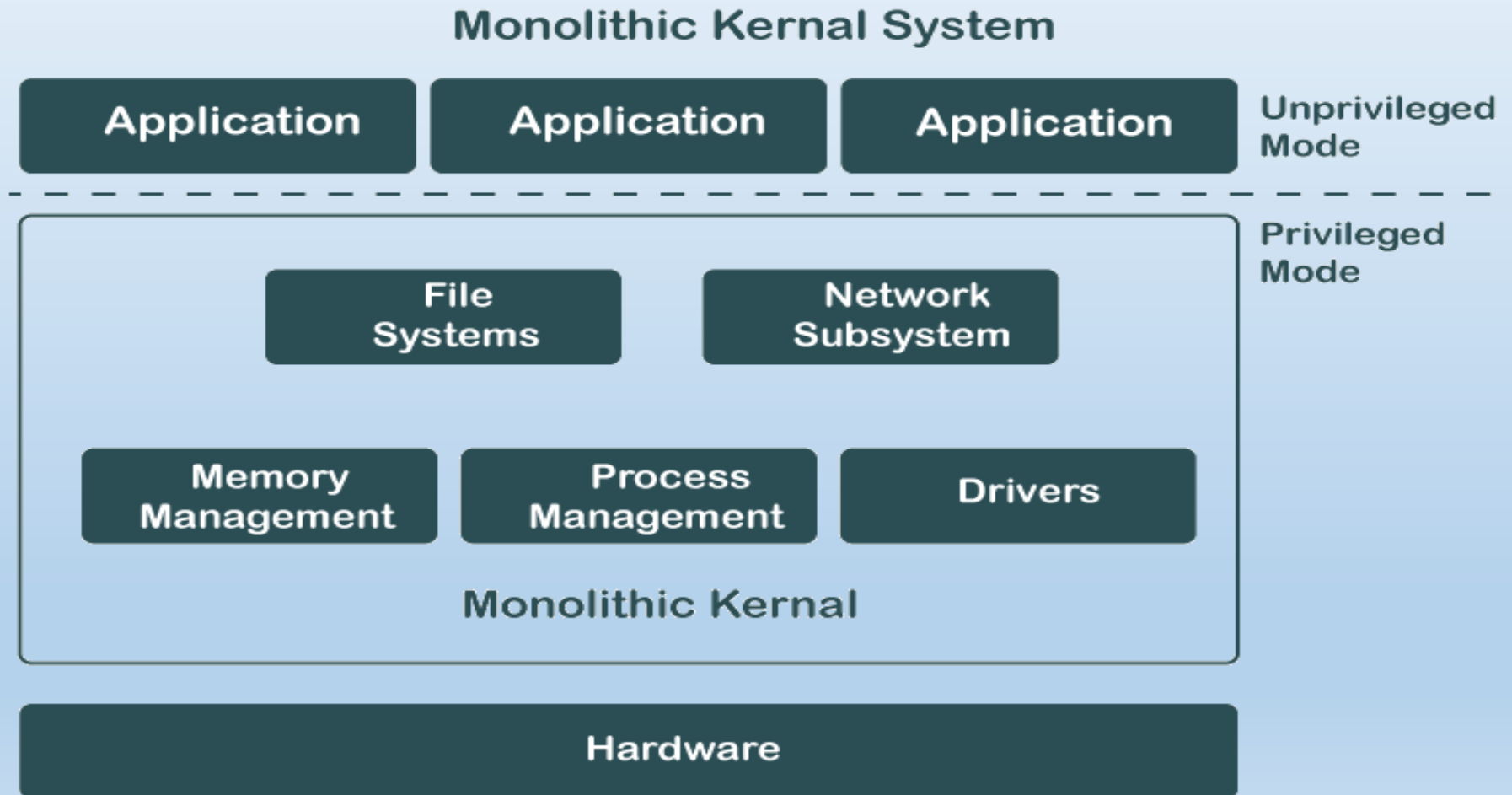
Operating System Structure

- ✓ We want a clear structure to let us apply an operating system to our particular needs because operating systems have complex structures.
- ✓ It is easier to create an operating system in pieces, much as we break down larger issues into smaller, more manageable sub-problems.
- ✓ Every segment is also a part of the operating system.
- ✓ Operating system structure can be thought of as the strategy for connecting and incorporating various operating system components within the kernel.
- ✓ Operating systems are implemented using many types of structures,

MONOLITHIC STRUCTURE

- ✓ The monolithic operating system controls all aspects of the operating system's operation, including file management, memory management, device management, and operational operations.
- ✓ The monolithic operating system is often referred to as the monolithic kernel.
- ✓ Multiple programming techniques such as batch processing and time-sharing increase a processor's usability.
- ✓ Working on top of the operating system and under complete command of all hardware, the monolithic kernel performs the role of a virtual computer.
- ✓ This is an old operating system that was used in banks to carry out simple tasks like batch processing and time-sharing, which allows numerous users at different terminals to access the Operating System.

The following diagram represents the monolithic structure:



Advantages

- ✓ Because layering is unnecessary and the kernel alone is responsible for managing all operations, it is easy to design and execute.
- ✓ Due to the fact that functions like memory management, file management, process scheduling, etc., are implemented in the same address area, the monolithic kernel runs rather quickly when compared to other systems. Utilizing the same address speeds up and reduces the time required for address allocation for new processes.

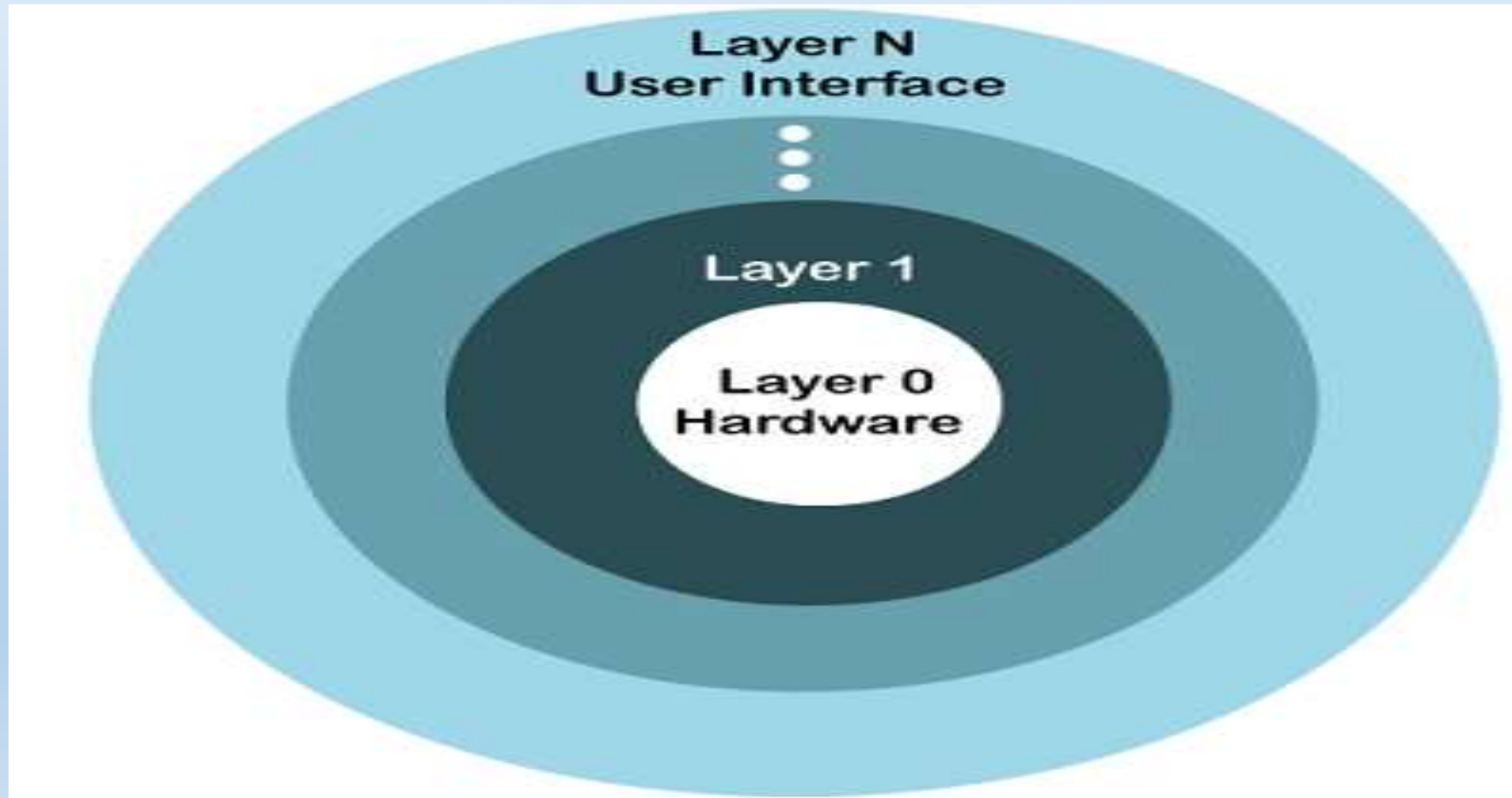
Disadvantages :

- ✓ The monolithic kernel's services are interconnected in address space and have an impact on one another, so if any of them malfunctions, the entire system does as well.
- ✓ It is not adaptable. Therefore, launching a new service is difficult.

LAYERED STRUCTURE

- ✓ The OS is separated into layers or levels in this kind of arrangement.
- ✓ Layer 0 (the lowest layer) contains the hardware, and layer 1 (the highest layer) contains the user interface (layer N).
- ✓ These layers are organized hierarchically, with the top-level layers making use of the capabilities of the lower-level ones.
- ✓ The functionalities of each layer are separated in this method, and abstraction is also an option.
- ✓ Because layered structures are hierarchical, debugging is simpler, therefore all lower-level layers are debugged before the upper layer is examined.
- ✓ As a result, the present layer alone has to be reviewed since all the lower layers have already been examined.

The image below shows how OS is organized into layers:



Advantages

- Work duties are separated since each layer has its own functionality, and there is some amount of abstraction.
- Debugging is simpler because the lower layers are examined first, followed by the top layers.

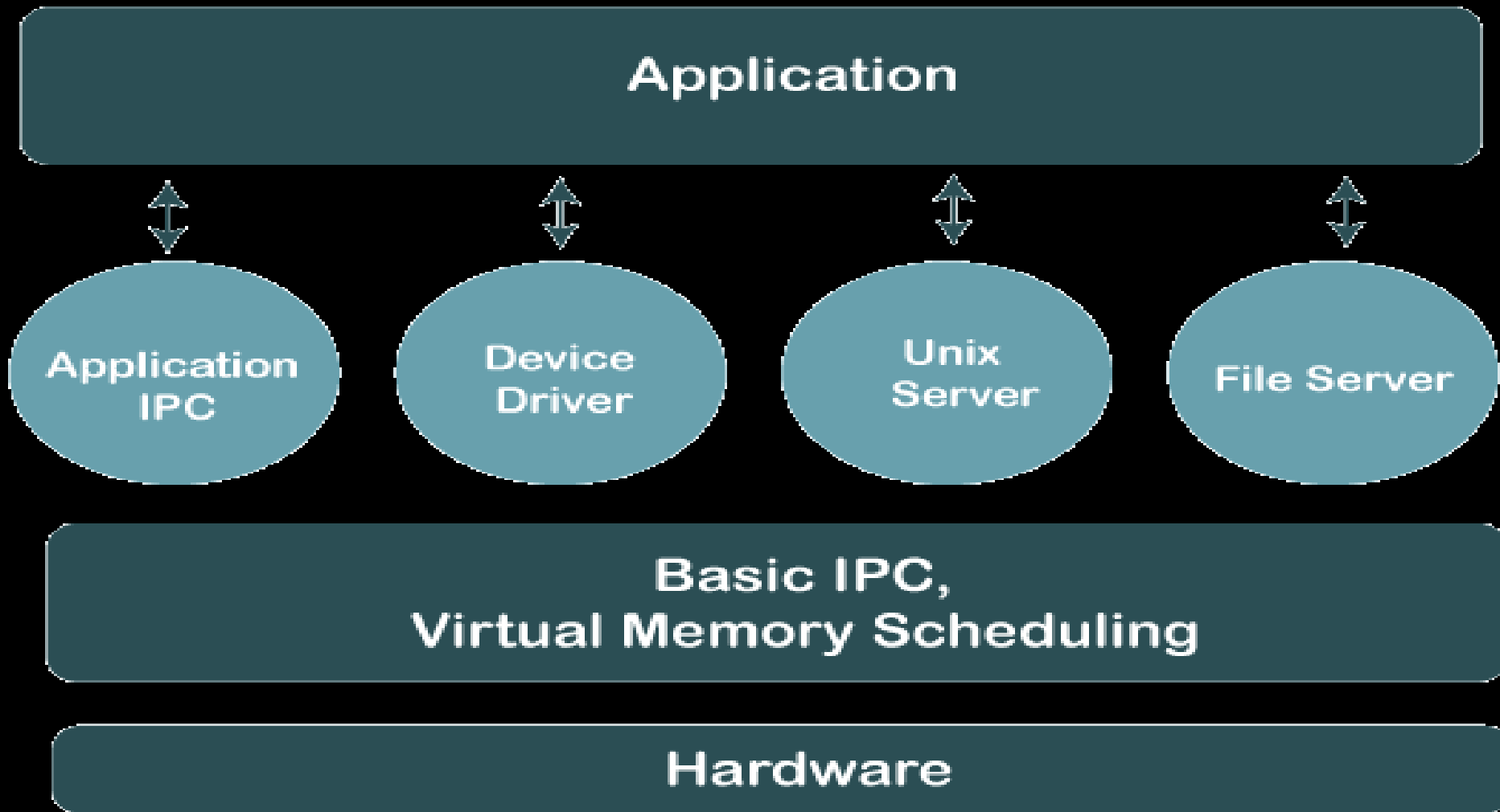
Disadvantages

- Performance is compromised in layered structures due to layering.
- Construction of the layers requires careful design because upper layers only make use of lower layers' capabilities.

MICRO-KERNEL STRUCTURE

- ✓ The operating system is created using a micro-kernel framework that strips the kernel of any unnecessary parts.
- ✓ Systems and user applications are used to implement these optional kernel components. So, Micro-Kernels is the name given to these systems that have been developed.
- ✓ Each Micro-Kernel is created separately and is kept apart from the others. As a result, the system is now more trustworthy and secure.
- ✓ If one Micro-Kernel malfunctions, the remaining operating system is unaffected and continues to function normally.

Microkernal Operating System



Advantages of Micro-Kernel Structure:

- ✓ It enables portability of the operating system across platforms.
- ✓ Due to the isolation of each Micro-Kernel, it is reliable and secure.
- ✓ The reduced size of Micro-Kernels allows for successful testing.
- ✓ The remaining operating system remains unaffected and keeps running properly even if a component or Micro-Kernel fails.

Disadvantages of Micro-Kernel Structure:

- ✓ The performance of the system is decreased by increased inter-module communication.
- ✓ The construction of a system is complicated.

EXOKERNEL

- ✓ An operating system called Exokernel was created at MIT with the goal of offering application-level management of hardware resources.
- ✓ The exokernel architecture's goal is to enable application-specific customization by separating resource management from protection. Exokernel size tends to be minimal due to its limited operability.
- ✓ Exokernel operating systems have a number of features, including:
 - Enhanced application control support.
 - Splits management and security apart.
 - A secure transfer of abstractions is made to an unreliable library operating system.
 - Brings up a low-level interface.
 - Operating systems for libraries provide compatibility and portability.

Advantages of Exokernel Structure:

- ✓ Application performance is enhanced by it.
- ✓ Accurate resource allocation and revocation enable more effective utilization of hardware resources.
- ✓ New operating systems can be tested and developed more easily.
- ✓ Every user-space program is permitted to utilise its own customised memory management.

Disadvantages of Exokernel Structure:

- ✓ A decline in consistency
- ✓ Exokernel interfaces have a complex architecture.

VIRTUAL MACHINES (VMs)

- ✓ The hardware of our personal computer, including the CPU, disc drives, RAM, and NIC (Network Interface Card), is abstracted by a virtual machine into a variety of various execution contexts based on our needs, giving us the impression that each execution environment is a separate computer.
- ✓ A virtual box is an example of it.
- ✓ Using CPU scheduling and virtual memory techniques, an operating system allows us to execute multiple processes simultaneously while giving the impression that each one is using a separate processor and virtual memory.
- ✓ System calls and a file system are examples of extra functionalities that a process can have that the hardware is unable to give.
- ✓ Instead of offering these extra features, the virtual machine method just offers an interface that is similar to that of the most fundamental hardware.
- ✓ A virtual duplicate of the computer system underneath is made available to each process.

Advantages of Virtual Machines:

- ✓ Due to total isolation between each virtual machine and every other virtual machine, there are no issues with security.
- ✓ A virtual machine may offer an architecture for the instruction set that is different from that of actual computers.
- ✓ Simple availability, accessibility, and recovery convenience.

Disadvantages of Virtual Machines:

- ✓ Depending on the workload, operating numerous virtual machines simultaneously on a host computer may have an adverse effect on one of them.
- ✓ When it comes to hardware access, virtual computers are less effective than physical ones.

Kernel

- ✓ Kernel is central component of an operating system that manages operations of computer and hardware.
- ✓ It basically manages operations of memory and CPU time. It is core component of an operating system.
- ✓ Kernel acts as a bridge between applications and data processing performed at hardware level using inter-process communication and system calls.
- ✓ Kernel loads first into memory when an operating system is loaded and remains into memory until operating system is shut down again.
- ✓ It is responsible for various tasks such as disk management, task management, and memory management.

Objectives of Kernel

To establish communication between user level application and hardware.

- ✓ To decide state of incoming processes.
- ✓ To control disk management.
- ✓ To control memory management.
- ✓ To control task management.

Types of Kernel :

1. Monolithic Kernel –

- ✓ It is one of types of kernel where all operating system services operate in kernel space. It has dependencies between systems components.
- ✓ It has huge lines of code which is complex.
- ✓ Example: Unix, Linux, Open VMS, XTS-400 etc.

Advantage:

- ✓ It has good performance.

Disadvantage:

- ✓ It has dependencies between system component and lines of code in millions.

2. Micro Kernel –

- ✓ It is kernel types which has minimalist approach. It has virtual memory and thread scheduling.
- ✓ It is more stable with less services in kernel space. It puts rest in user space.
- ✓ Example : Mach, L4, AmigaOS, Minix, K42 etc.

Advantage :

- ✓ It is more stable.

Disadvantage :

- ✓ There are lots of system calls and context switches.

3. Hybrid Kernel –

- ✓ It is the combination of both monolithic kernel and microkernel.
- ✓ It has speed and design of monolithic kernel and modularity and stability of microkernel.
- ✓ Example : Windows NT, Netware, BeOS etc.

Advantage :

- ✓ It combines both monolithic kernel and microkernel.

Disadvantage :

- ✓ It is still similar to monolithic kernel.

4. Exo- Kernel –

- ✓ It is the type of kernel which follows end-to-end principle. It has fewest hardware abstractions as possible. It allocates physical resources to applications.
- ✓ Example : Nemesis, Ex-OS etc.

Advantage :

- ✓ It has fewest hardware abstractions.

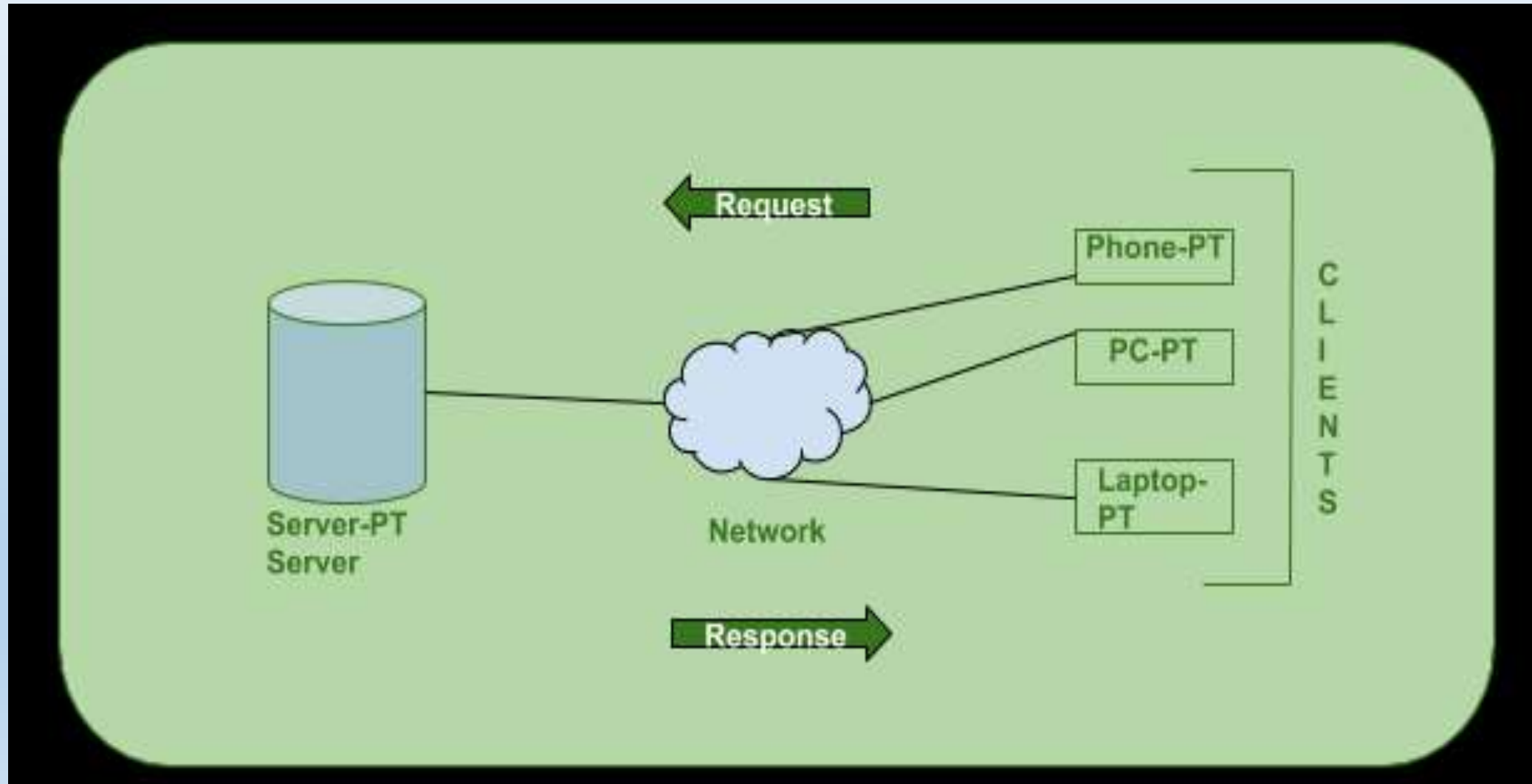
Disadvantage :

- ✓ There is more work for application developers.

Client-Server Model

- ✓ The Client-server model is a distributed application structure that partitions task or workload between the providers of a resource or service, called servers, and service requesters called clients.
- ✓ In the client-server architecture, when the client computer sends a request for data to the server through the internet, the server accepts the requested process and deliver the data packets requested back to the client.
- ✓ Clients do not share any of their resources.
- ✓ Examples of Client-Server Model are Email, World Wide Web, etc.

Client-Server Model



Advantages of Client-Server model:

- ✓ Centralized system with all data in a single place.
- ✓ Cost efficient requires less maintenance cost and Data recovery is possible.
- ✓ The capacity of the Client and Servers can be changed separately.

Disadvantages of Client-Server model:

- ✓ Clients are prone to viruses, Trojans and worms if present in the Server or uploaded into the Server.
- ✓ Server are prone to Denial of Service (DOS) attacks.
- ✓ Data packets may be spoofed or modified during transmission.

What Is a Shell?

- A shell is a program that takes commands typed by the user and calls the operating system to run those commands.
- A shell is a program that acts as the interface between you and the Linux system, allowing you to enter commands for the operating system to execute.
- Shell accepts your instruction or commands in English and translate it into computers native binary language